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Requirements Document

Pakistan Disaster Management System

Course: Software Construction & Development **Version:** 0.1 (this document will be updated as the project evolves, following an Agile approach)

1. Group Members

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2. Problem Description

Pakistan is one of the countries most affected by natural disasters. Every year people face floods, heavy rain, heatwaves, landslides, snowfall, smog and earthquakes. The 2022 floods affected millions of people, and in 2025 flash floods in Buner and Swat caused heavy loss of life.

Weather and disaster information is available, but it is spread across many different websites and apps. Most of it is technical and hard for ordinary people to understand. People often do not know if their own district is in danger, how serious the danger is, or what they should do to stay safe.

Our project solves this by bringing live weather data, earthquake data, automatic warnings, past disaster history and safety precautions together in one simple web application with a map.

3. Objectives

1. Show live weather forecasts (current and 7 days) for districts across Pakistan.
2. Show recent earthquakes in and near Pakistan.
3. Automatically generate disaster warnings with severity levels (moderate, high, severe) using simple rules.
4. Show warnings and hazards on an interactive map so users can see danger areas easily.
5. Remind users of past disasters in their area so they take warnings seriously.
6. Give clear safety precautions and emergency numbers for each type of disaster.
7. Allow admins to post official alerts and add incidents.

4. Intended Users

General Public: Ordinary people in Pakistan who want to check weather and danger warnings for their district. They can view the map and warnings without logging in, and can register to save their home district.

Registered Users: Users who create an account and choose a home district, so they can quickly see warnings for their own area.

Admins: Trusted people (for example, the project team acting as an authority) who can post official alerts and add incidents on the map.

5. Functional Requirements

Functional requirements describe what the system must do.

Weather and Earthquakes

- FR1: The system shall fetch current weather and a 7-day forecast for Pakistani districts from the Open-Meteo API.
- FR2: The system shall fetch recent earthquakes in and around Pakistan from the USGS Earthquake API.
- FR3: The system shall store the list of districts with their names and map coordinates in the database.

Warnings

- FR4: The system shall automatically check weather and earthquake data against threshold rules written in Python.
- FR5: The system shall generate warnings for heavy rain, flood, landslide, heatwave, snow, strong wind, earthquake and smog.
- FR6: The system shall give each warning a severity level: moderate, high or severe.

Map

- FR7: The system shall display districts, warnings and earthquakes on an interactive map of Pakistan.
- FR8: The system shall use a different icon or emoji for each type of hazard.
- FR9: The system shall show warning details when a user clicks a map marker.

Disaster History

- FR10: The system shall store records of past disasters (such as the 2005 earthquake, 2022 floods and 2025 Buner/Swat flash floods).
- FR11: The system shall show a history note with a warning when that area had a similar disaster before.

Precautions

- FR12: The system shall show precautions for before, during and after each type of disaster.
- FR13: The system shall show emergency numbers: Rescue 1122, Police 15, Fire 16 and Edhi 115.

Users

- FR14: The system shall allow users to register with a username and password.
- FR15: The system shall allow registered users to log in and log out.
- FR16: The system shall allow users to choose and change their home district.

Admin

- FR17: The system shall allow admins to add incidents on the map.
- FR18: The system shall allow admins to post official alerts.
- FR19: The system shall prevent normal users from opening admin pages.

6. Non-Functional Requirements

Non-functional requirements describe how well the system should work.

- NFR1 (Usability): The interface shall be simple, use clear language, and use a dark theme.
- NFR2 (Performance): Pages shall load within about 5 seconds on a normal internet connection.
- NFR3 (Security): User passwords shall be stored in hashed form, never as plain text.
- NFR4 (Reliability): If an external API fails, the system shall show an error message instead of crashing.
- NFR5 (Maintainability): The code shall be written mainly in Python, divided into small functions with clear names and comments.
- NFR6 (Portability): The system shall run on a normal Windows computer using free tools only.
- NFR7 (Cost): The system shall use only free APIs that do not need an API key.
- NFR8 (Testability): Important Python functions, such as the warning rules, shall be tested using pytest.

7. Prioritized Feature List (Product Backlog)

ID	Feature	Priority	Planned Version
F1	Requirements document and repository setup	High	v0.1.0
F2	Flask project setup and district list	High	v0.2.0
F3	Live weather forecast (current + 7 days)	High	v0.2.0
F4	Interactive map with district markers	High	v0.3.0
F5	Automatic warning rules with severity levels	High	v0.3.0
F6	Recent earthquakes from USGS	High	v0.4.0
F7	Past disaster history notes	Medium	v0.4.0
F8	Precautions and emergency numbers	Medium	v0.4.0
F9	User registration, login and home district	Medium	v0.5.0
F10	Admin panel (incidents and official alerts)	Medium	v0.5.0
F11	Air quality and smog warnings	Low	v0.6.0
F12	Automated tests with pytest	Medium	v0.6.0
F13	Dark theme and final design improvements	Low	v1.0.0

This backlog will be updated during the project. New features may be added and priorities may change based on progress and feedback.

8. Assumptions

- The free Open-Meteo and USGS APIs will remain available and free during the project.
- Users will have an internet connection to see live data.
- Weather forecasts from the APIs are accurate enough for basic warnings.
- Users can use a basic web browser.

9. Constraints

- The project must be written mainly in Python, using Flask and SQLite.
- Only free tools and free APIs can be used.
- JavaScript must be kept to the minimum needed for the map.
- The project must be finished within the semester deadline.
- Team members have limited programming experience, so features are kept simple and built step by step.

10. Known Limitations

- This is a student project, not an official government system. Warnings are based on simple rules and may not be as accurate as official NDMA or PMD warnings.
- Flood warnings are estimated from rainfall only, not from real river levels.
- The system does not send SMS or mobile notifications.
- Only selected major districts will be included, not every district in Pakistan.
- Disaster history data is entered manually and covers only major past events.
- The system will run on a local computer and will not be hosted online during development.